

Principles of Quantum Mechanics

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1 Introduction

Recall from IB Quantum Mechanics we have the following:

- (i) Wave-particle duality. Waves behave like particles, like light quanta, and vice versa (particles behave like waves).
- (ii) We have a wavefunction $\psi(x)$ for a particle which we interpret as a probability density $|\psi(x)|^2$ (i.e. probability is intrinsic to the theory).
- (iii) Observables become operators on wavefunctions:
 - $\hat{x} \rightarrow x$;
 - $\hat{p} \rightarrow -i\hbar \frac{\partial}{\partial x}$

The lack of commutation, since $[\hat{x}, \hat{p}] = i\hbar$, leads to the uncertainty relation.

- (iv) Schrödinger's equation with Hamiltonian H specifies dynamics, and eigenvalues of H are the allowed energy levels of the system.
- (v) All of this explains the hydrogen atom structure and the resulting line spectra.

The aims of this course are the following:

- (i) Reformulate quantum mechanics in a more powerful, flexible and useful form: the Dirac formulation, which goes beyond wavefunctions. The previous description appears as a particular realisation of our generalisation of quantum states and also leads to alternatives such as

$$\tilde{\psi}(p) \quad \text{with} \quad \hat{x} \rightarrow i\hbar \frac{\partial}{\partial p} \quad \text{and} \quad \hat{p} \rightarrow p.$$

- (ii) This more general approach allows much simpler analysis of known problems (for example, the Simple Harmonic Oscillator), but also lots of new things such as *spin* of particles and symmetries (translations, rotations) and the related conservation laws.
- (iii) We will also look at identical particles (bosons and fermions).
- (iv) We will develop a framework for quantising more general systems such as EM fields.
- (v) We will also set up perturbation theory, which is a practical tool for computation, but we will not heavily consider the detailed applications.

2 Dirac Formalism

2.1 Concepts and Definitions

A quantum system is described at each instant by a *state* $|\psi\rangle$ which belongs to a complex vector space V ; any non-zero element of this space V is a possible state. Thus

$$|\psi\rangle, |\phi\rangle \in V \implies \alpha |\psi\rangle + \beta |\phi\rangle \in V \quad \text{for any } \alpha, \beta \in \mathbb{C}.$$

Physically, this is a superposition principle leading to wave-like behaviour, but these states are *not wavefunctions*. There are also *dual* or conjugate states $\langle\psi|$ which belong to the dual space

$V^\dagger$. By definition, states and duals can be combined or paired to give a complex number. The map

$$(\langle\psi|, |\phi\rangle) \mapsto \langle\psi|\phi\rangle$$

defines a map from $V^\dagger \times V \rightarrow \mathbb{C}$ with

$$\langle\phi|(\alpha_1|\psi_1\rangle + \alpha_2|\psi_2\rangle) = \alpha_1\langle\phi|\psi_1\rangle + \alpha_2\langle\phi|\psi_2\rangle$$

and

$$(\beta_1\langle\phi_1| + \beta_2\langle\phi_2|)|\psi\rangle = \beta_1\langle\phi_1|\psi\rangle + \beta_2\langle\phi_2|\psi\rangle.$$

The spaces V and $V^\dagger$ come with an inner product which can be described as a one-to-one correspondence between the spaces

$$\begin{aligned} V &\rightarrow V^\dagger \\ |\psi\rangle &\rightarrow \langle\psi| = (|\psi\rangle)^\dagger. \end{aligned}$$

We also have that if we take a linear combination of states to the dual space, the coefficients have to have their complex conjugates taken when mapping from the dual space to V or vice versa. Hence the inner product is

$$|\phi\rangle, |\psi\rangle \mapsto \langle\phi|\psi\rangle = (|\phi\rangle)^\dagger |\psi\rangle$$

and it is assumed to obey

$$\langle\phi|\psi\rangle = (\langle\psi|\phi\rangle)^*$$

since the inner product is Hermitian, and the inner product defines a positive-definite norm on V :

$$\| |\psi\rangle \|^2 = \langle\psi|\psi\rangle.$$

Remark. Knowing $\langle\phi|\psi\rangle$ for all $\langle\phi|$ determines $|\psi\rangle$ and vice versa.

We can show this by

$$\langle\phi|\psi_1\rangle = \langle\phi|\psi_2\rangle \implies \langle\phi|\chi\rangle = 0,$$

where $|\chi\rangle = |\psi_1\rangle - |\psi_2\rangle$. If this holds for all $\langle\phi|$, we can set $\langle\phi| = \langle\chi|$, so by positive definiteness we have that the norm of $|\chi\rangle$ is zero, and therefore $|\chi\rangle = 0$ too.

We also have that the physical content of any state is unchanged by

$$|\psi\rangle \rightarrow \alpha |\psi\rangle$$

for any non-zero $\alpha \in \mathbb{C}$. So we will usually normalise to make the state have norm 1, but we can still change it by a phase factor.

Remark. Relative phases can be important in expressions where we're adding a linear combination of elements of V . We also assume the space V is complete so appropriate sequences and series converge in V .

A complete inner product space is called a *Hilbert space*.

Definition. (Operator) An *operator* Q is a linear map on states

$$|\psi\rangle \rightarrow Q|\psi\rangle.$$

The same operator can be thought of as acting to the left on dual states, so

$$\langle\phi| \rightarrow \langle\phi| Q$$

where the right-hand side is defined by

$$(\langle\phi| Q)|\psi\rangle = \langle\phi|(Q|\psi\rangle) = \langle\phi| Q|\psi\rangle.$$